

MBA, 2nd semester

Studiengang / Fachsemester Study program / Subject semester

Analytical skills for Business.

Prüfungsfach / Prüfungsdozent:in Examination subject / Examiner

① Version Control Systems

Git:

- A distributed version control system for managing code, tracking changes, and facilitating collaborative work space for modern development environments.
- GUI: Command Line interface (by default)
- works without Internet on local machine
- Collaborative tools only via email

Git-Hub:

- Git-Hub is cloud based platform for hosting Git repositories on cloud enabling to work from anywhere.
- GUI: website / Desktop app.
- Requires Internet
- provides social features, collaborative tools, & ~~per~~ (pull requests, track issues, invite others to collaborate).
- provides project management tools like CI/CD via GitHub actions.
- Host static webpages via GitHub pages.

The primary purpose of Git is to setup a VCS on the local machine.

The primary purpose of GitHub is to host the project to cloud.

(b).

Git-repository: A storage ~~loc~~ location for all project files, with complete track of history of changes and modifications.

Commit: A snapshot of project files in a particular time with complete track of history

Branch: provides a dedicated line development to work on different aspects/features on the same project.

Importance in Collaborative analytics projects.

- keeps track of all analytical steps and data processing steps.
- Two or more analysts can work on same or different features at a time facilitating collaborative research.
- Data Governance.
- protection against data loss.

Git

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(2). Python.

Here is an incomplete list of primary use cases in business analytics.

- Data analysis and manipulation, (e.g. EDA, Descriptive analysis, predictive analysis etc.)
- python facilitates ML/AI via packages and libraries which can be used for predictive analysis.
- python can also be used to REST API's to integrate and communicate with other software applications.

Eg: A business can predict sales of Q3 ~~using~~ using ML algo trained on past data.

SQL.

Here is an incomplete list of primary use cases in business analytics.

~~Query~~

- Querying large datasets
- To understand what the data is and how the data structure is by using simple SQL queries.
- ~~To connect~~ To connect SQL database to other tools like uku for automation process.

Eg: A business can use simple query to know to total sales in Q3 instead ~~data~~ downloading and performing analysis using R/Python.

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(b). Libraries:

A library is a pre-defined code or algorithm which can perform a specific action or multiple tasks.

example:

tidyverse is a package in R which is used to perform tasks related data analysis and manipulation.

lm() is a package in R which can be only used for ~~regression~~ simple regression analysis.

Libraries can reduce time & effort in the data analysis process.

Functions:

A set of rules and definitions, script that can be reused anywhere in the project.

eg: By creating a function to load data and summarise the data, the same function can be used anywhere else in the same project file/files to perform the same steps for a different dataset.

Variables:

Variables facilitate to store values.

eg: Code in R.

$X \leftarrow c(3, 5, 7, 9, 80)$

here the values are stored in X.

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Variables are useful in data analysis to store values in different formats (like dataframes, vector) to reuse/call and update in the further analysis process.

(c). Docker is a Containerization platform/service which encapsulates the

- Application code
- system configurations
- runtime
- required libraries & packages

into a container.

Thus the main benefit of a container is its portability. Once a project or analytical steps are fit into a container, this container can be hosted anywhere and used by other team members. It makes the system look less complex and organised.

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(d). IDE

Integrated Development environment is an software application which provides .

- Code editor
- debugger
- Collaboration tools via VCS .

eg: VScode, Rstudio .

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(3) (a).

n8n.

- ↳ no-code workflow automation and integration tool.
- ↳ provides visual boxes to connect to different services (via apis), perform automation, integrate with other tools/services.

QGIS.

- ↳ geographic information system
- ↳ used to analyse geographic data, spatial data, location data.

Power BI

- ↳ Data analysis, reporting, and business intelligence tool, developed by microsoft.
- ↳ strongly integrates with MS tools like excel.

Snowflake

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- ↳ is a cloud data warehouse for large datasets.

(b). Advantages of using no-code/low-code tools

- Reproducibility & automation
- Advanced features (via ML/AI, apis)
- Integration with other platforms (eg. via apis) & Collaboration
- easy learning curve.
- scalable

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e.g. excel might not be a best choice for large datasets and collaborative work instead users can use tools like Airtable with nfu for Collaboration and automation work.

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(4). Central tendency.

- (a).
- mean is the arithmetic average and used to know the average value in sample of data (eg. avg. marks in class).
 - Median is the middle value, can be used what middle values look like in a given dataset.
 - mode is the most frequent value.
eg: can be used most frequent marks students scored.

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(5). Univariate

- to know the relationship of one variable.
- generally perform central tendency, sd, variance.

Bivariate

- ~~to~~ to know the relationship between two variable.
- generally perform correlation analysis, regression analysis.

- multivariate Data

- To know the relationship between multiple variable

- Complex analysis, PCA.

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(5).

$$(a) \quad 3 + \sqrt{9} - 2^{\{2\}} \times 4/3.$$

(b). getwd().

(c). ~~?~~ mean().

(d). sales_data <- %>%

values?

Region <- c("North", "South", "East")

Sales <- c(45000, 38000, 52000).

)

print(sales_data).

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4.(c).

Missing values.

- Completely missing values

- MCAR, MAR.

Strategies.

- Completely remove.

- use mean, median, mode

- use algos like KNN.

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Inconsistent Data

- Dates format.

- Variables like New York as "NY"

- Time format

Strategies.

- Text normalization

- Standardization.

- manual fix.

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